AI-POWERED STUDENT PERFORMANCE PREDICTION & LEARNING RECOMMENDATION SYSTEM

AI & ML Major Project | 10-Day Industry-Oriented Assignment

Technology stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Streamlit and Joblib

Dataset: Synthetic educational demonstration data (no real student records)

Prepared as a reproducible learning prototype

Abstract

This project demonstrates an end-to-end machine-learning workflow to estimate a broad student performance category from study habits and academic indicators. A Streamlit application accepts inputs, shows an estimated category and class probabilities, and provides transparent rule-based learning suggestions. The dataset is synthetic; results are for educational demonstration only.

1. Problem Statement

Academic support teams may review attendance, study hours and assessment indicators manually. This prototype explores whether these variables can be combined into an estimated performance category. Predictions are estimates, not definitive judgments.

2. Objectives

Prepare data and handle missing values; explore distributions and relationships; train and compare a baseline with a Random Forest classifier; evaluate on held-out test data; build a Streamlit dashboard; and document limitations and responsible use.

3. Dataset Description

The dataset contains 1200 synthetic records and seven input features: study hours per week, attendance percentage, previous exam marks, assignment completion percentage, sleep hours per day, internet access and class participation. The target contains Needs Support, Developing and On Track. Missing values were introduced in selected input columns to demonstrate preprocessing. Labels were generated from a noisy formula based on the features, not collected from real students.

4. Methodology

The target was kept separate from the input features. Data was split into 80% training and 20% test sets using stratification and random_state=42. Numerical features use median imputation and standard scaling; categorical features use most-frequent imputation and one-hot encoding. All transformations are fitted inside a scikit-learn pipeline on training data. A Random Forest classifier with 250 trees was compared with a majority-class DummyClassifier.

5. Exploratory Data Analysis

The app provides a data preview, category counts and numerical correlations. The reports folder includes a performance-category distribution chart and a confusion matrix. Because the data is synthetic, patterns are properties of the data generator and must not be presented as findings about real students.

6. Model Evaluation

On the held-out synthetic test set, Random Forest accuracy was 0.613, macro-F1 was 0.507, and majority-class baseline accuracy was 0.542. The class-level report and confusion matrix are included in the project files. These metrics demonstrate the workflow only; they do not validate real-world performance.

Metric	Result
Random Forest test accuracy	0.613
Random Forest macro-F1	0.507
Majority baseline accuracy	0.542

7. Application Modules

The Streamlit app has three tabs: (1) interactive prediction and study suggestions, (2) dataset exploration, and (3) model methodology and limitations. The model predicts a category; a separate set of explicit rules generates suggestions such as improving attendance, practising regularly, completing assignments and maintaining a

consistent sleep schedule.

8. Limitations and Ethics

Synthetic data cannot represent the complexity of real educational outcomes. The target is derived from a simulated formula and reflects its assumptions. Predictions can be wrong. Do not use this prototype for grading, admissions, discipline or any high-stakes decision. Do not upload identifiable student records. Real deployment would require consent, privacy safeguards, representative data, expert review, fairness assessment and monitoring.

9. Conclusion

The prototype demonstrates data preparation, exploratory analysis, model training, held-out evaluation and deployment through Streamlit. The synthetic dataset makes the workflow reproducible for a student demonstration. Future work could evaluate a legally shareable anonymized public dataset under appropriate ethical safeguards.

10. Tools and References

Python; Pandas; NumPy; Scikit-learn; Matplotlib; Streamlit; Joblib. Dataset: generated synthetically for this educational project; no external student dataset was used.